

Mathematics Stage 3 -- Year B -- Unit 22 Addition & Subtraction Strategies and Decimals

KLA: Mathematics | Stage 3 | Year 6 | Term: _____ | Duration: 2-3 weeks | Teacher: _____ | Year: _____

Syllabus outcomes

Code	Outcome descriptor
MAO-WM-01	Develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly.
MA3-AR-01	Selects and applies appropriate strategies to solve addition and subtraction problems.
MA3-RN-01	Applies an understanding of place value and the role of zero to represent the properties of numbers.
MA3-RN-02	Compares and orders decimals up to 3 decimal places.

Syllabus content points (addressed through SOLO tracker)

All content points below are covered by students working independently through the SOLO tracker. Mini masterclasses target specific outcomes based on live tracker data.

Red -- Foundation	Yellow -- Applying	Green -- Extension
Decimal place value (RN-A) + estimation & multi-number addition (AR-A) Express thousandths as decimals; interpret decimal notation; indicate the place value of digits to 3 dp; partition decimals using place value (RN-A). Place decimals to 3 dp on a number line; approximate the size of decimals (RN-A). Round numbers appropriately when estimating; use place value to check calculations for errors (AR-A). Use place value to add 3 or more numbers with different numbers of digits (AR-A).	Efficient strategies, multistep problems & decimal operations (applying) Choose, compare, evaluate and communicate efficient strategies for addition and subtraction (AR-B). Solve multistep word problems, including problems that require more than one operation (AR-B). Model the addition and subtraction of decimals up to 3 decimal places (AR-B). Solve decimal add/subtract word problems and justify why the strategy is appropriate (AR-B).	Extended -- Stage 4 Decimals, Integers & Order of Operations Multiply and divide decimals by powers of 10; round decimals to a specified number of places (MA4-FRC-C-01). Compare, order, add and subtract positive and negative integers (MA4-INT-C-01). Apply the order of operations to evaluate expressions (MA4-INT-C-01).

Teaching model

Students work independently through the Word Labs SOLO Tracker (wordlabs.app/solo) across differentiated bands -- Grow (resources), Know (practice), Show (auto-marked assessments). The teacher monitors live dashboard data and pulls small groups for targeted mini masterclasses (15-40 min) based on which outcomes have the most incomplete Show results. Remaining session time is used for incidental one-on-one and small group teaching in response to student questions and tracker data.

Working Mathematically (MAO-WM-01)

Process	How it appears in this unit
Communicating	Students explain and compare addition and subtraction strategies, read and write decimals to thousandths, and justify why a strategy is appropriate.
Understanding and fluency	Students interpret decimal place value, place decimals on a number line, add 3+ numbers, round to estimate, and add/subtract decimals to 3 dp.
Reasoning	Students reason about which strategy is most efficient, use place value to check for errors, and judge whether an answer is reasonable.
Problem solving	Students solve multistep and decimal word problems, and (Stage 4) work with decimal multiplication/division, integers and the order of operations.

Differentiation

Support	HPGE / Extension
Students in this group:	Students in this group:

Pre-test

Date	
Format	
Band place ments / key findings	

Post-test / unit evaluation

Date	
Key findings	
Students who need further consolidation	
Next steps	

Resources

See Appendix for Resources

Word Labs SOLO Tracker: wordlabs.app/solo | NSW DoE Stage 3 Year B Unit 22 | NESA Mathematics K-10 Syllabus (2022)

Unit 22 -- Outcome Teaching Record

Find the outcome/s you taught, write your notes, sign and date. Order follows the outcome sequence, not your teaching calendar.

Co de	Outcome	Syllabus content point/s addressed	Mini masterclass teaching notes	Sign	Date
R1	Decimal place value to thousandths.	RN-A: express thousandths as decimals; interpret decimal notation; indicate place value to 3 dp; partition decimals MA3-RN-01 Year A	Mini Lesson 1		
R2	Place decimals on a number line and approximate.	RN-A: place decimals to 3 dp on a number line; approximate the size of decimals MA3-RN-02 Year A	Mini Lesson 2		
R3	Round to estimate and check calculations.	AR-A: round numbers when estimating; use place value to check for errors in calculations MA3-AR-01 Year A	Mini Lesson 3		
R4	Add three or more numbers using place value.	AR-A: use place value to add or subtract 3 or more numbers with different numbers of digits MA3-AR-01 Year A	Mini Lesson 4		
Y1	Choose, compare and explain efficient strategies.	AR-B: choose and use efficient strategies; compare, evaluate and communicate strategies MA3-AR-01 Year B	Mini Lesson 5		
Y2	Solve multistep word problems.	AR-B: solve multistep word problems, including problems requiring more than one operation MA3-AR-01 Year B	Mini Lesson 6		
Y3	Add and subtract decimals to 3 dp.	AR-B: model the addition and subtraction of decimals up to 3 dp using appropriate representations MA3-AR-01 Year B	Mini Lesson 7		
Y4	Solve decimal word problems and justify the strategy.	AR-B: solve decimal add/subtract word problems; justify why the strategy is appropriate MA3-AR-01 Year B	Mini Lesson 8		
G1	Multiply and divide decimals.	Stage 4 FRC: multiply and divide decimals by powers of 10; round to a specified number of places MA4-FRC-C-01 Stage 4 (Core)	Mini Lesson 9		
G2	Add and subtract integers.	Stage 4 INT: compare and order integers; add and subtract positive and negative integers MA4-INT-C-01 Stage 4 (Core)	Mini Lesson 10		
G3	Order of operations.	Stage 4 INT: apply the order of operations to evaluate expressions MA4-INT-C-01 Stage 4 (Core)	Mini Lesson 11		

Syllabus outcomes and content from Mathematics K-10 Syllabus (2022) copyright NSW Education Standards Authority (NESA) for and on behalf of the Crown in right of the State of New South Wales, 2024. AR = Additive relations. RN = Represents numbers. A/B = syllabus content group (Year A ~ Year 5; Year B ~ Year 6).

Unit 22 -- Mini Lesson Sequence

Addition & Subtraction Strategies and Decimals | Stage 3 Year B

Supplement to the SOLO Teacher Program (pages 1-2). Print pages 1-2 only for annotation. Reference these lesson plans when planning mini masterclasses or when a reviewer requires evidence of planned lesson content.

Red band -- Foundation -- Mini Lessons 1-4

Yellow band -- On-level -- Mini Lessons 5-8

Green band -- Extension -- Mini Lessons 9-11

DoE lesson links: Lessons marked DoE Lesson X draw directly from NSW DoE Mathematics Stage 3 Unit 22. Steps have been condensed to 20-40 minutes for mini masterclass delivery. Resources referenced by their original resource numbers from the DoE unit.

Mini Lesson 1 Decimal place value to thousandths **[HANDS-ON]**

R1 | Red band | DoE Lesson 5 (adapted)

Duration: 25-30 min | **Outcomes:** R1 | Syllabus: MA3-RN-01, MAO-WM-01 | RN-A: thousandths; place value; partition decimals

Linked to: DoE Lesson 5 (adapted) -- Lesson 5 -- Decimal place value | Key resources: Place value charts (to thousandths), decimal expander cards

Physical materials needed: place value charts, decimal expander cards

Learning intention

Students are learning to:
read, interpret and partition decimals to thousandths using place value

Success criteria

Students can:

- * I can express thousandths as decimals (e.g. 7 thousandths = 0.007).
- * I can say the place value of each digit in a decimal to 3 dp.
- * I can partition a decimal using place value (e.g. $0.346 = 0.3 + 0.04 + 0.006$).

Mini masterclass structure (25-30 min)

Activate (4 min): Where is the thousandths place in 0.346? What does the 6 stand for?

Model (10 min): On a place value chart show tenths, hundredths, thousandths. $0.346 = 3$ tenths + 4 hundredths + 6 thousandths. Write 7 thousandths as 0.007.

Guided practice (10 min): Students name the place value of digits and partition decimals to 3 dp.

Check (4 min): Exit question: partition 0.582 and say what the 2 stands for.

Key vocabulary

decimal, tenths, hundredths, thousandths, place value, partition, digit

Differentiation

Support: Use a place value chart so each digit sits in the right column.
Extension: Write a decimal in expanded form using fractions (e.g. $3/10 + 4/100 + 6/1000$).

Assessment

Students interpret and partition decimals — links to R1 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 2 Decimals on a number line [HANDS-ON]

R2 | Red band | DoE Lesson 5 (adapted)

Duration: 25 min | **Outcomes:** R2 | Syllabus: MA3-RN-02, MAO-WM-01 | RN-A: decimals on a number line; approximate

Linked to: DoE Lesson 5 (adapted) -- Lesson 5 -- Decimal place value | Key resources: Decimal number lines

Physical materials needed: decimal number lines

Learning intention	Success criteria
Students are learning to: place decimals to 3 dp on a number line and approximate their size	Students can: * I can place a decimal to 3 dp on a number line between two whole numbers. * I can split an interval into tenths and hundredths to locate a decimal. * I can approximate the size of a decimal (e.g. 0.62 is about 0.6).

Mini masterclass structure (25 min)

Activate (3 min): Where would 0.6 go on a line from 0 to 1?

Model (10 min): Divide 0 to 1 into tenths; place 0.6. Zoom into 0.6-0.7 in hundredths to place 0.62. Approximate: 0.62 is close to 0.6.

Guided practice (8 min): Students place decimals on number lines and approximate their size.

Check (4 min): Exit question: place 0.45 on a number line and say which tenth it is closest to.

Key vocabulary	Differentiation
decimal, number line, interval, tenths, hundredths, approximate, between	Support: Provide a pre-marked number line in tenths. Extension: Place three decimals to 3 dp on the same line and order them.
Assessment	Teaching notes (annotate here)
Students place and approximate decimals — links to R2 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 3 Rounding to estimate and check

R3 | Red band | DoE Lesson 2 (adapted)

Duration: 25-30 min

Outcomes: R3

Syllabus: MA3-AR-01, MAO-WM-01 | AR-A: round to estimate; check for errors

Linked to: DoE Lesson 2 (adapted) -- Lesson 2 -- Estimate and check the reasonableness of a solution | Key resources: Rounding number lines

Learning intention

Students are learning to:
round numbers to estimate and use place value to check calculations

Success criteria

Students can:

- * I can round numbers to the nearest 10, 100 or 1000 to make an estimate.
- * I can estimate a sum or difference before calculating.
- * I can use place value to spot an error in a calculation.

Mini masterclass structure (25-30 min)

Activate (4 min): Roughly, what is $412 + 389$? Estimate before adding.

Model (10 min): Round $412 \rightarrow 400$ and $389 \rightarrow 400$, so the sum is about 800 (exact 801). If someone got 80 or 8010, place value shows it is wrong.

Guided practice (10 min): Students estimate sums/differences and check given answers for place-value errors.

Check (4 min): Exit question: estimate $596 + 207$. Is 803 reasonable?

Key vocabulary

round, estimate, nearest, reasonable, place value, check, error

Differentiation

Support: Provide a number line to round on.

Extension: Estimate a 3-number sum and explain how close the estimate is.

Assessment

Students round to estimate and check — links to R3 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 4 Adding three or more numbers

R4 | Red band | DoE Lesson 3 (adapted)

Duration: 25 min | **Outcomes:** R4 | Syllabus: MA3-AR-01, MAO-WM-01 | AR-A: add 3+ numbers with place value

Linked to: DoE Lesson 3 (adapted) -- Lesson 3 -- Solving problems with efficient strategies | Key resources: Place value charts

Learning intention		Success criteria
Students are learning to: use place value to add 3 or more numbers with different numbers of digits		Students can: * I can line up numbers by place value before adding. * I can add 3 or more numbers with different numbers of digits. * I can regroup (carry) when a column totals 10 or more.
Mini masterclass structure (25 min)		
Activate (3 min): How would you set out $248 + 19 + 1305$ to add them? Model (10 min): Line up ones under ones, tens under tens, etc. Add each column, regrouping when needed: $248 + 19 + 1305 = 1572$. Guided practice (8 min): Students add 3 or more numbers with different numbers of digits using place value. Check (4 min): Exit question: $56 + 408 + 9 = ?$		
Key vocabulary		Differentiation
place value, column, regroup, carry, sum, line up, digit		Support: Use a place value chart / grid paper so columns line up. Extension: Add four numbers including a 4-digit number, then estimate to check.
Assessment		Teaching notes (annotate here)
Students add 3+ numbers using place value — links to R4 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 5 Choosing and comparing efficient strategies

Y1 | Yellow band | DoE Lesson 4 (adapted)

Duration: 30 min | **Outcomes:** Y1 | Syllabus: MA3-AR-01, MAO-WM-01 | AR-B: choose, compare and evaluate strategies

Linked to: DoE Lesson 4 (adapted) -- Lesson 4 -- Comparing and evaluating strategies | Key resources: Strategy cards (jump, split, compensation, bridging)

Learning intention		Success criteria	
Students are learning to: choose an efficient strategy and compare which works best		Students can: * I can use strategies like bridging and compensation for addition and subtraction. * I can choose an efficient strategy for a given problem. * I can compare two strategies and explain which is better.	
Mini masterclass structure (30 min)			
Activate (4 min): What is $199 + 47$? Is there a quicker way than column addition?			
Model (12 min): Compensation: $199 + 47 = 200 + 47 - 1 = 246$. Bridging: $47 + 199 \rightarrow 47 + 200$ then back 1. Compare which is more efficient and why.			
Guided practice (10 min): Students solve problems two ways and decide which strategy is most efficient.			
Check (4 min): Exit question: solve $502 - 48$ using an efficient strategy and explain your choice.			
Key vocabulary		Differentiation	
strategy, efficient, bridging, compensation, jump, split, evaluate, compare		Support: Provide a menu of strategies to choose from. Extension: Invent a problem where compensation is clearly the best strategy.	
Assessment		Teaching notes (annotate here)	
Students choose and compare strategies — links to Y1 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 6 Multistep word problems

Y2 | Yellow band | DoE Lesson 1 (adapted)

Duration: 30 min | **Outcomes:** Y2 | Syllabus: MA3-AR-01, MAO-WM-01 | AR-B: multistep word problems

Linked to: DoE Lesson 1 (adapted) -- Lesson 1 -- Solving multistep word problems | Key resources: Word problem cards

Learning intention		Success criteria	
Students are learning to: solve multistep word problems requiring more than one operation		Students can: * I can break a problem into steps and decide which operation each step needs. * I can carry out the steps in order to reach the answer. * I can check my answer is reasonable.	
Mini masterclass structure (30 min)			
Activate (4 min): Read a 2-step problem aloud. Ask: what do we work out first? Model (12 min): Model a multistep problem: buy 3 items at \$4 then pay with \$20 — first multiply ($3 \times 4 = 12$), then subtract ($20 - 12 = 8$). Guided practice (10 min): Students solve multistep word problems, showing each step. Check (4 min): Exit question: a 2-step problem with addition then subtraction.			
Key vocabulary		Differentiation	
multistep, operation, step, multiply, add, subtract, estimate, reasonable		Support: Provide a step-by-step planning frame (Step 1 ..., Step 2 ...). Extension: Write a multistep problem of your own and solve it.	
Assessment		Teaching notes (annotate here)	
Students solve multistep word problems — links to Y2 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 7 Adding and subtracting decimals [HANDS-ON]

Y3 | Yellow band | DoE Lesson 6 & 7 (adapted)

Duration: 30 min | **Outcomes:** Y3 | Syllabus: MA3-AR-01, MAO-WM-01 | AR-B: add/subtract decimals to 3 dp

Linked to: DoE Lesson 6 & 7 (adapted) -- Lesson 6 -- Adding decimals; Lesson 7 -- Subtracting decimals | Key resources: Place value charts, number lines

Physical materials needed: place value charts, number lines

Learning intention	Success criteria
Students are learning to: add and subtract decimals up to 3 decimal places	Students can: * I can line up the decimal points to add or subtract decimals. * I can use zeros as place holders so both numbers have the same decimal places. * I can add and subtract decimals to 3 dp.

Mini masterclass structure (30 min)

Activate (4 min): Add \$3.50 and \$2.75. How do you set it out?

Model (12 min): Line up the decimal points; add place by place (\$6.25). Pad zeros: $3.25 + 1.4 = 3.250 + 1.400 = 4.650$. Subtract: $7.5 - 3.25 = 4.25$.

Guided practice (10 min): Students add and subtract decimals to 3 dp, lining up the points.

Check (4 min): Exit question: $6.4 - 2.75 = ?$

Key vocabulary	Differentiation
decimal, add, subtract, line up, decimal point, place holder, place value	Support: Use a place value chart so points and columns line up. Extension: Solve a 3-decimal calculation that needs regrouping across the decimal point.
Assessment	Teaching notes (annotate here)
Students add and subtract decimals — links to Y3 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 8 Decimal word problems and justifying

Y4 | Yellow band | DoE Lesson 8 (adapted)

Duration: 30 min **Outcomes:** Y4 Syllabus: MA3-AR-01, MAO-WM-01 | AR-B: decimal word problems; justify

Linked to: DoE Lesson 8 (adapted) -- Lesson 8 -- Decimal problem solving | Key resources: Money/measurement problem cards (AUD)

Learning intention		Success criteria
Students are learning to: solve decimal word problems and justify why the strategy is appropriate		Students can: * I can solve word problems by adding and subtracting decimals to 3 dp. * I can choose an appropriate strategy for the problem. * I can justify why my strategy is effective.
Mini masterclass structure (30 min)		
Activate (4 min): You spend \$7.45 and \$2.80. How much change from \$20? Model (12 min): Add $7.45 + 2.80 = 10.25$, then $20 - 10.25 = 9.75$. Explain why adding first, then subtracting, is the efficient order. Guided practice (10 min): Students solve decimal word problems and write a sentence justifying their strategy. Check (4 min): Exit question: a 2-step money problem; justify your method.		
Key vocabulary		Differentiation
decimal, word problem, strategy, justify, efficient, change, total, AUD		Support: Provide a sentence starter for the justification ('I chose ... because ...'). Extension: Solve a measurement decimal problem (e.g. lengths in metres) and justify.
Assessment		Teaching notes (annotate here)
Students solve and justify decimal problems — links to Y4 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 9 Multiplying and dividing decimals

G1 | Green band | Original mini lesson (Stage 4 extension)

Duration: 30 min | **Outcomes:** G1 | Syllabus: MA4-FRC-C-01, MAO-WM-01 | Stage 4: $\times/\div$ decimals by powers of 10; round

Learning intention		Success criteria	
Students are learning to: multiply and divide decimals by powers of 10 and round to a given number of places		Students can: * I can multiply a decimal by 10, 100 and 1000 (shift digits left). * I can divide a decimal by 10, 100 and 1000 (shift digits right). * I can round a decimal to 1 or 2 decimal places.	
Mini masterclass structure (30 min)			
Activate (3 min): What is 3.45×100 ? Predict the digit shift. Model (12 min): $\times 100$ shifts two places left ($3.45 \rightarrow 345$). $\div 10$ shifts one place right ($3.45 \rightarrow 0.345$). Round 3.456 to 2 dp $\rightarrow 3.46$. Guided practice (10 min): Students multiply and divide decimals by powers of 10 and round. Check (4 min): Exit question: $0.7 \times 1000 = ?$ and round 2.358 to 1 dp.			
Key vocabulary		Differentiation	
decimal, multiply, divide, powers of 10, round, decimal place, shift		Support: Use a place value chart and move every digit the same number of places. Extension: Divide a decimal by 1000 and round the result to 2 dp.	
Assessment		Teaching notes (annotate here)	
Students multiply/divide decimals and round — extends R1 and Y3.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 10 Adding and subtracting integers

G2 | Green band | Original mini lesson (Stage 4 extension)

Duration: 30 min **Outcomes:** G2 Syllabus: MA4-INT-C-01, MAO-WM-01 | Stage 4: compare/order/add/subtract integers

Learning intention		Success criteria	
Students are learning to: compare, order, add and subtract positive and negative integers		Students can: * I can compare and order integers using < and >. * I can add a negative integer (move left on the number line). * I can subtract to cross zero into negative numbers.	
Mini masterclass structure (30 min)			
Activate (4 min): Which is greater, −5 or −2? Use a number line to decide. Model (12 min): On a number line: $3 + (-5) = -2$ (move left 5); $-1 - 3 = -4$. Order a set like −4, 2, −1, 3. Guided practice (10 min): Students compare, order, add and subtract positive and negative integers. Check (4 min): Exit question: work out $-3 + 5$ and $2 - 6$.			
Key vocabulary		Differentiation	
integer, positive, negative, less than, greater than, add, subtract, number line		Support: Use a number line for every calculation to make direction concrete. Extension: Solve a real-life integer problem (temperature change or money owed).	
Assessment		Teaching notes (annotate here)	
Students work with integers — extends R3 and R4.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 11 Order of operations G3 Green band Original mini lesson (Stage 4 extension)	
Duration: 30 min	Outcomes: G3 Syllabus: MA4-INT-C-01, MAO-WM-01 Stage 4: order of operations
Learning intention	Success criteria
Students are learning to: use the order of operations to evaluate expressions	Students can: * I know we agree on an order so everyone gets the same answer. * I can do brackets first, then multiplication and division, then addition and subtraction. * I can evaluate expressions with and without brackets.
Mini masterclass structure (30 min)	
Activate (4 min): Work out $2 + 3 \times 4$ two different ways. Why do we need an agreed order? Model (12 min): Order of operations: brackets, then $\times$ and $\div$, then $+$ and $-$. So $2 + 3 \times 4 = 14$, but $(2 + 3) \times 4 = 20$ — brackets change the answer. Guided practice (10 min): Students evaluate expressions with and without brackets. Check (4 min): Exit question: work out $10 - (2 + 3) \times 1$ and $20 \div (5 - 1)$.	
Key vocabulary	Differentiation
order of operations, brackets, grouping symbols, evaluate, expression	Support: Provide the operation order as a checklist to follow. Extension: Place brackets in a number sentence to make it equal a given target.
Assessment	Teaching notes (annotate here)
Students apply the order of operations — extends Y2.	Taught as described / adapted / replaced -- note here:

Unit 22 -- Resource Appendix

Every link verified. Videos open in a browser; worksheets are print-ready PDFs (see the download checklist at the end).

Type	Resource	Link
R1 -- Decimal place value to thousandths.		
Video	Place Value to the Thousandths — Representing Decimal Numbers Miacademy	https://www.youtube.com/watch?v=BsZJGMRWEK0
Video	Expanded Form with Decimals Math with Mr. J	https://www.youtube.com/watch?v=GX8o-S6Vxig
Worksheet (print)	Corbettmaths — Place Value (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf
R2 -- Place decimals on a number line and approximate.		
Video	Writing Tenths and Hundredths (Decimals and Fractions) Math with Mr. J	https://www.youtube.com/watch?v=ibR_iBxnITE
Video	Comparing and Ordering Decimals Math with Mr. J	https://www.youtube.com/watch?v=KWslYjXKhrM
Worksheet (print)	Corbettmaths — Ordering Decimals (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/ordering-decimals-pdf.pdf
R3 -- Round to estimate and check calculations.		
Video	Estimating Whole Number Sums and Differences Math with Mr. J	https://www.youtube.com/watch?v=EvQf38lnAJc
Video	Estimating Decimal Sums Math with Mr. J	https://www.youtube.com/watch?v=x05LHZJNKQ0
Worksheet (print)	Corbettmaths — Rounding (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/rounding-pdf.pdf
Worksheet (print)	Corbettmaths — Estimation (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/estimation-pdf.pdf
R4 -- Add three or more numbers using place value.		
Video	How to Add and Subtract Large Numbers Math with Mr. J	https://www.youtube.com/watch?v=186wdnoAYJc
Video	Adding Large Numbers — A Quick Review Math with Mr. J	https://www.youtube.com/watch?v=T9asFgfN5bg
Worksheet (print)	Corbettmaths — Place Value (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf
Y1 -- Choose, compare and explain efficient strategies.		
Video	How to Use Bridging to Support Mental Addition and Subtraction Third Space Learning	https://www.youtube.com/watch?v=oDDC1thyc2Q
Video	Addition and Subtraction — The Compensation Strategy Charlotte Dooley	https://www.youtube.com/watch?v=X3kgzlb8VrM
Worksheet (print)	Corbettmaths — Place Value (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf
Y2 -- Solve multistep word problems.		
Video	Multi-Step Word Problem (Addition, Subtraction, Multiplication) Khan Academy	https://www.youtube.com/watch?v=-sSDb_wZqKQ
Video	How to Add and Subtract Large Numbers Math with Mr. J	https://www.youtube.com/watch?v=186wdnoAYJc
Worksheet (print)	Corbettmaths — Estimation (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/estimation-pdf.pdf
Y3 -- Add and subtract decimals to 3 dp.		
Video	Adding and Subtracting Decimals (How to) Math with Mr. J	https://www.youtube.com/watch?v=PnwLv6khwk8
Video	How to Add and Subtract Decimals (Step-by-Step Examples) Math with Mr. J	https://www.youtube.com/watch?v=Cg-_TeiaSa8
Worksheet (print)	Corbettmaths — Adding Decimals (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/adding-decimals-pdf.pdf
Worksheet (print)	Corbettmaths — Subtracting Decimals (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/subtracting-decimals-pdf.pdf
Y4 -- Solve decimal word problems and justify the strategy.		
Video	How to Add and Subtract Decimals (Step-by-Step Examples) Math with Mr. J	https://www.youtube.com/watch?v=Cg-_TeiaSa8
Video	Estimating Decimal Sums Math with Mr. J	https://www.youtube.com/watch?v=x05LHZJNKQ0

Type	Resource	Link
Worksheet (print)	Corbettmaths — Adding Decimals (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/adding-decimals-pdf.pdf
G1 -- Multiply and divide decimals.		
Video	Multiply a Whole Number by a Decimal Math with Mr. J	https://www.youtube.com/watch?v=tsOibhsgYoQ
Video	How to Multiply a Whole Number and a Decimal Math with Mr. J	https://www.youtube.com/watch?v=FbZr5br9ZOg
Worksheet (print)	Corbettmaths — Multiplying and Dividing Decimals (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/decimals-multiplying-and-dividing.pdf
G2 -- Add and subtract integers.		
Video	How to Add Negative Numbers (Integers) Math with Mr. J	https://www.youtube.com/watch?v=1VFj_coqKQU
Video	Subtracting Integers Using a Number Line Math with Mr. J	https://www.youtube.com/watch?v=Dfytkh_IYME
Worksheet (print)	Corbettmaths — Negative Numbers (PDF)	https://corbettmaths.com/wp-content/uploads/2018/11/Negative-Numbers.pdf
G3 -- Order of operations.		
Video	Order of Operations — A Step-By-Step Guide PEMDAS Math with Mr. J	https://www.youtube.com/watch?v=x41BCrYh8Kc
Video	Order of Operations — Parentheses, Brackets and Braces PEMDAS Math with Mr. J	https://www.youtube.com/watch?v=u7Lack4Tmx8
Worksheet (print)	Corbettmaths — Order of Operations (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/order-of-operations-pdf.pdf
Beyond SOLO -- Stage 4 extension		
Video	Multiply a Whole Number by a Decimal Math with Mr. J	https://www.youtube.com/watch?v=tsOibhsgYoQ
Video	How to Add Negative Numbers (Integers) Math with Mr. J	https://www.youtube.com/watch?v=1VFj_coqKQU
Video	Order of Operations — A Step-By-Step Guide PEMDAS Math with Mr. J	https://www.youtube.com/watch?v=x41BCrYh8Kc
Worksheet (print)	Corbettmaths — Multiplying and Dividing Decimals (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/decimals-multiplying-and-dividing.pdf
Worksheet (print)	Corbettmaths — Negative Numbers (PDF)	https://corbettmaths.com/wp-content/uploads/2018/11/Negative-Numbers.pdf

Worksheet download checklist

Download and print these PDFs before the unit (or save to your class drive). Tick each as you go.

- ☐ Corbettmaths — Place Value (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf>
- ☐ Corbettmaths — Ordering Decimals (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/ordering-decimals-pdf.pdf>
- ☐ Corbettmaths — Rounding (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/rounding-pdf.pdf>
- ☐ Corbettmaths — Estimation (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/estimation-pdf.pdf>
- ☐ Corbettmaths — Adding Decimals (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/adding-decimals-pdf.pdf>
- ☐ Corbettmaths — Subtracting Decimals (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/subtracting-decimals-pdf.pdf>
- ☐ Corbettmaths — Multiplying and Dividing Decimals (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/decimals-multiplying-and-dividing.pdf>
- ☐ Corbettmaths — Negative Numbers (PDF) -- <https://corbettmaths.com/wp-content/uploads/2018/11/Negative-Numbers.pdf>
- ☐ Corbettmaths — Order of Operations (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/order-of-operations-pdf.pdf>